

WEST PENOBSCOT BAY BUOY 44033, ME, USA

WMO: 997180

Lat: 44.055N

Lon: 68.996W

Elev: 0

StdP: 101.33

Time Zone: -5.00 (NAE)

Period: 05-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-13.4	-11.0	N/A	N/A	N/A	N/A	N/A	N/A	16.5	-4.8	15.2	-3.1	7.6	300	0.666

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.2	21.0	N/A	19.9	N/A	19.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	3.7	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.0	11.8	10.7	DB	-16.8	24.4	2.4	1.6	-18.5	25.6	-19.9	26.6	-21.2	27.5	-22.9	28.7	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.8	-2.3	-2.2	0.6	5.6	9.7	13.6	16.6	16.9	15.2	11.1	6.8	1.5		
	DBStd	7.52	5.12	4.08	3.55	2.26	2.21	2.06	1.93	1.30	1.95	2.67	3.63	4.54		
	HDD10.0	1571	381	342	292	131	33	1	0	0	0	20	106	265		
	HDD18.3	3857	639	575	551	381	268	143	61	47	97	224	347	523		
	CDD10.0	767	0	0	0	1	23	108	203	215	154	55	8	0		
	CDD18.3	10	0	0	0	0	0	0	6	3	2	0	0	0		
	CDH23.3	4	0	0	0	0	0	0	3	1	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	5.3	7.0	6.6	6.4	5.2	4.4	3.8	3.3	3.3	4.3	5.9	6.3	6.9		
(15) Precipitation	PrecAvg	1288	105	89	113	116	103	104	86	77	113	134	134	125		
	PrecMax	1856	194	204	213	251	251	248	186	179	253	399	236	223		
	PrecMin	768	13	31	28	9	18	52	36	21	34	32	65	49		
	PrecStd	240	46	36	47	60	52	53	42	42	63	78	39	40		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.2	6.6	8.5	12.7	18.9	20.4	23.2	21.9	21.6	17.8	14.0	11.2		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	7.6	5.2	6.5	10.4	15.4	18.5	21.5	20.3	19.5	15.8	12.9	10.1		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	6.4	4.3	5.5	9.2	13.8	17.2	20.1	19.4	18.4	15.1	12.1	8.9		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	5.1	3.4	4.7	8.4	12.6	16.3	19.1	18.8	17.6	14.4	11.6	7.9		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.1	5.5	4.7	3.6	3.5	3.5	3.7	3.2	3.2	3.8	4.2	5.4		
		MCDBR	6.1	5.4	4.9	5.0	5.6	5.4	5.4	4.2	4.0	3.8	3.9	6.0		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.299	0.307	0.329	0.355	0.381	0.423	0.442	0.411	0.368	0.339	0.320	0.299		
	taud		2.473	2.464	2.464	2.425	2.395	2.293	2.262	2.337	2.439	2.518	2.535	2.516		
	Ebn at Noon		833	890	912	910	890	850	828	843	857	841	802	798		
	Edh at Noon		71	85	97	109	116	129	132	118	99	80	67	63		
(44) All-Sky Solar Radiation	RadAvg		1.59	2.48	3.69	4.62	5.24	5.65	5.92	5.33	4.29	2.75	1.71	1.26		
	RadStd		0.12	0.20	0.23	0.49	0.63	0.47	0.43	0.28	0.26	0.25	0.15	0.13		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+60	N/A

Nomenclature: See separate page